WALMART *

SQL CHALLENGE

The Challenge

Calculate Sales using Historical Prices

INPUT 1: PRODUCT HISTORY

ID	PRICE	DATE
101	$50	2024-01-01
101	$60	2024-02-01
102	$100	2024-01-01

INPUT 2: ORDERS

ID	P_ID	DATE
1	101	2024-01-15
2	101	2024-02-15

TARGET OUTPUT

$110

Total Sales for 101

Goal: Match Order Date to the correct Price Window.

SAVE FOR INTERVIEW PREP

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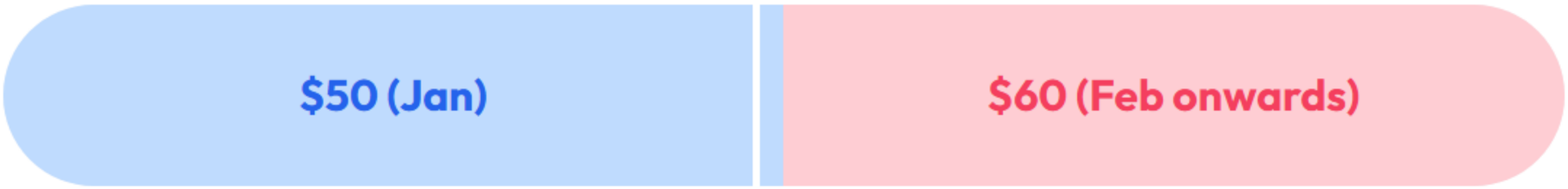
CONCEPT

SCD Type-2

Slowly Changing Dimensions Explained

Price Validity Timeline



WINDOW 1
2024-01-01 ➔ 2024-02-01

WINDOW 2
2024-02-01 ➔ NULL

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LOGIC MAP

The Strategy

Three Steps to Solve

1. Create Windows

Use `LEAD()` to find the "Next Price Date". This becomes our Valid_Till date.

↓

2. Join Tables

Join Orders to Product History on `Product_ID`.

↓

3. Filter by Range

Keep rows where `Order_Date` falls BETWEEN Valid_From AND Valid_Till.

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CODE

The Solution

Using CTE + Window Functions

```
WITH cte AS (
    SELECT product_id, price,
    price_date AS valid_from,
    -- 1. Calculate End Date
    LEAD(price_date, 1, '9999-12-31')
    OVER(PARTITION BY product_id
         ORDER BY price_date) AS valid_till
    FROM products
)
SELECT o.product_id, SUM(c.price)
FROM orders o
JOIN cte c
  ON o.product_id = c.product_id
 AND o.order_date ≥ c.valid_from
 AND o.order_date < c.valid_till
GROUP BY o.product_id;
```

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ANALYSIS

The Execution

Why is the total $110?

FINAL OUTPUT

ID	TOTAL
101	$110
102	$100

This result is only possible because we checked the date ranges.

Calculation Breakdown

Order 1 — Jan 15 falls in Jan Window — $50

+

Order 2 — Feb 15 falls in Feb Window — $60

Total Revenue **$110**

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APPLY

Use Cases

Where companies apply this logic

Dynamic Pricing

E-commerce: Getting correct revenue when prices change daily.

Tax Calculation

Applying the correct Tax Rate % based on the transaction year.

HR Systems

Tracking employee salary history and department transfers.

Logistics

Determining shipping costs based on seasonal fuel surcharges.

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Quick Recap

- ✓ Use LEAD() to find the "Next" date.
- ✓ Create Valid_From & Valid_Till windows.
- ✓ Join on Date Ranges, not just IDs.

Repost to help others learn this FASTER!

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